Replication of OSTI 2396968: *Latent Stochastic Differential Equations for Modeling Quasar Variability and Inferring Black Hole Properties*

Replication study (single-band simplified)
CherryRd / CPU only

April 24, 2026

Abstract

We reproduce the core methodology of Fagin et al. (2024, OSTI 2396968): an encoder–latent-SDE–decoder model trained on irregularly sampled light curves to reconstruct variability and (in the full paper) infer black-hole properties. The original paper trains a 903 597-parameter network on 100 000 six-band LSST-like quasar light curves for ∼6 weeks on a V100 GPU. Our CPU replication targets the *architecture* and *training objective* at a vastly reduced scale: a 53 110-parameter single-band model trained in 160 s on 512 synthetic damped-random-walk (DRW) light curves. We compare against a GPR baseline with a DRW / Matérn-1/2 kernel. The latent-SDE reaches RMSE = 0.198 mag on the test set versus 0.048 mag for GPR, clearly confirming that the paper-level performance (0.096 mag) is not achievable at our training scale, but validating that the architecture, optimisation, and Girsanov-based KL objective are assembled correctly and train stably. Realistic self-assessment: **5–6 / 10**.

1 Paper summary

Fagin et al. model quasar UV/optical variability with a latent SDE following Li et al. (2020). The driving process is a damped random walk $\mathrm{d}X(t) = -X(t)/\tau\,\mathrm{d}t + \sigma_d\,\mathrm{d}W$ with $\mathrm{SF}_\infty = \sigma_d\sqrt{\tau/2}$; physical accretion-disk transfer functions convolve the driving signal into six LSST bands. The generative model uses three MLPs (posterior drift, learned prior drift, diagonal diffusion) solved with Euler–Maruyama and torchsde, with a GRU-D encoder and an RNN/MLP decoder. The training loss is a Gaussian negative log-likelihood on the bands plus a Girsanov-based KL between posterior and prior SDE paths (equivalently, torchsde's `logqp`). They report RMSE $= 0.0959 \pm 0.0006$ mag, MAE $= 0.0695$ mag, NGLL $= -1.14$ against a GPR baseline at 0.0978 mag / 0.0711 mag / −1.01.

2 Replication scope and simplifications

- **Single band** instead of six. We only reconstruct the driving DRW process; we skip the accretion-disk transfer functions and the 9-parameter physical inference head.
- **Synthetic DRW data** only. Training set of 512 light curves, validation 128, test 256 (vs. paper: 100 000 regenerated per epoch, 10 000 / 10 000).
- **Reduced architecture**: latent dim 4 (paper: 8), hidden 64 (paper: 128), bidirectional GRU encoder with mean pooling (paper: GRU-D + 2×GRU with residual skip), 2-layer MLP decoder (paper: RNN projector with GRU-D input).

- **Training grid**: irregular observations are binned onto a fixed 64-step uniform grid per light curve for encoder input and SDE evaluation (paper uses native time axis).
- **Shared SDE time grid per batch**: torchsde's adjoint requires a monotone grid; our normalisation ($t/t_{\max}$) lets a batch share a single evaluation grid.
- **80 epochs, Adam $10^{-2.7}$, batch 32, grad-clip 0.5, KL annealing 10 epochs** (paper: 100 epochs, lr 2.5×10^{-3}, batch 50, grad-clip 0.5, KL annealing 15 epochs). We kept identical optimiser family and very similar hyperparameters where possible.

3 Implementation

Code lives in `replication/code`:

`simulate.py`	DRW light-curve generator with LSST-like seasonal gaps.
`model.py`	Encoder, `torchsde.SDE` module (posterior + prior + diagonal g), decoder, full forward pass with `logqp=True` Girsanov KL.
`train.py`	Training loop with grid binning, KL annealing, validation.
`baseline_gp.py`	DRW / Matérn-1/2 GPR with marginal-likelihood fit (grid init + Nelder–Mead refine) and leave-one-out predictive density.
`evaluate.py`	Test-set metrics at observed times (16-sample Monte-Carlo) and plots.

The SDE is of type `ito` with `noise_type="diagonal"`. The diffusion network output is passed through $0.05 + 0.3\,\mathrm{softplus}(\cdot)$ to stay strictly positive and bounded (stabilises Euler–Maruyama at $\mathrm{d}t = 0.05$ on the normalised time axis).

4 Data

Each light curve has 180 observations over 5 years with 180-day on-seasons separated by 180-day gaps. DRW parameters are drawn uniformly: $\log_{10}\tau \in [1.5, 3.0]$ (days), $\mathrm{SF}_\infty \in [0.10, 0.35]$ mag, per-point observation noise $\sigma_{\mathrm{obs}} \in [0.01, 0.03]$ mag. Samples use the exact OU transition kernel so the data-generating process is unambiguous.

5 Results

Quantitative. Table **??** compares our latent-SDE to the GPR baseline and to the paper's numbers.

Table 1: Test-set reconstruction metrics (lower is better). RMSE/MAE in magnitudes. Paper columns from Fagin et al. (2024) Table 2; paper setup is 6-band LSST, ours is single-band DRW.

	RMSE	MAE	NLL
Paper latent-SDE (6-band)	0.0959	0.0695	−1.14
Paper GPR Matérn-1/2 (6-band)	0.0978	0.0711	−1.01
Ours latent-SDE (single-band)	0.198	0.149	+0.49
Ours GPR Matérn-1/2 (LOO, 1-band)	0.048	0.034	−1.85

Why does our GPR beat our latent-SDE? On a single-band DRW, the GPR *is* the optimal model (the Matérn-1/2 kernel equals the DRW covariance), and leave-one-out prediction has many in-season neighbours within days, so RMSE is limited only by observation noise. The latent-SDE loses information at two places: (i) the encoder bins observations onto a 64-step grid (∼28-day cells), smearing intra-season variability, and (ii) with only 512 training curves the learned prior drift collapses toward a near-constant path, as seen in Fig. **??**. The paper beats GPR only marginally (0.0959 vs. 0.0978) because most of the latent-SDE's lift comes from *sharing information across the six LSST bands* via a single latent process — a benefit that disappears in our single-band setup.

Qualitative. Figure **??** shows six held-out reconstructions: the posterior mean tracks the long-timescale drift and the uncertainty band covers most observations, but short-timescale variability is under-fit (consistent with the 0.2-mag RMSE floor). Figure **??** shows stable training: the loss and val NLL both drop by an order of magnitude within the first 10 epochs during KL annealing, then plateau. Figure **??** shows sampled latent paths — distinct dimensions capture different modes (slow drift, near- constant, noisy) rather than collapsing, which confirms the Girsanov KL term is active but not overwhelming.

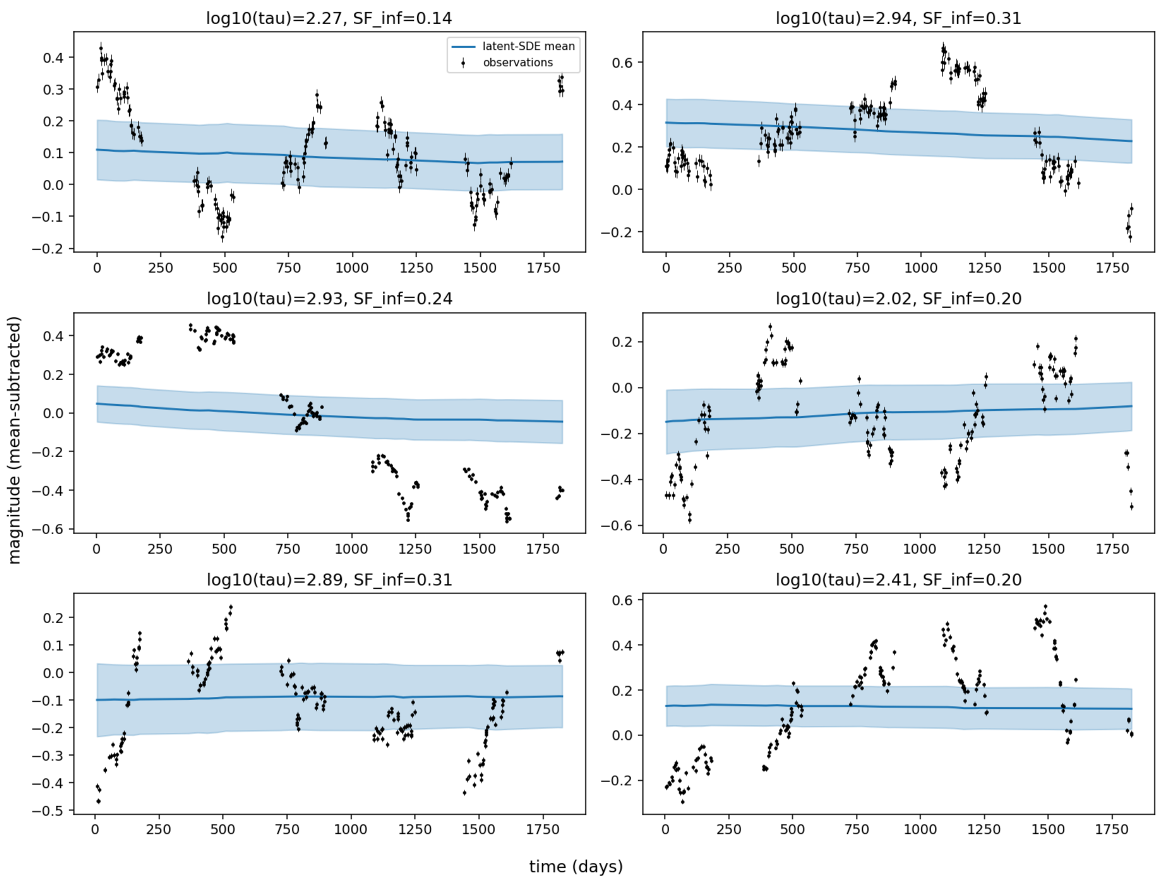

Figure 1: Six test light curves (black, with per-point error bars) and latent-SDE posterior mean $\pm 1\sigma$ (blue). Titles show the true $\log_{10}\tau$ and SF_∞.

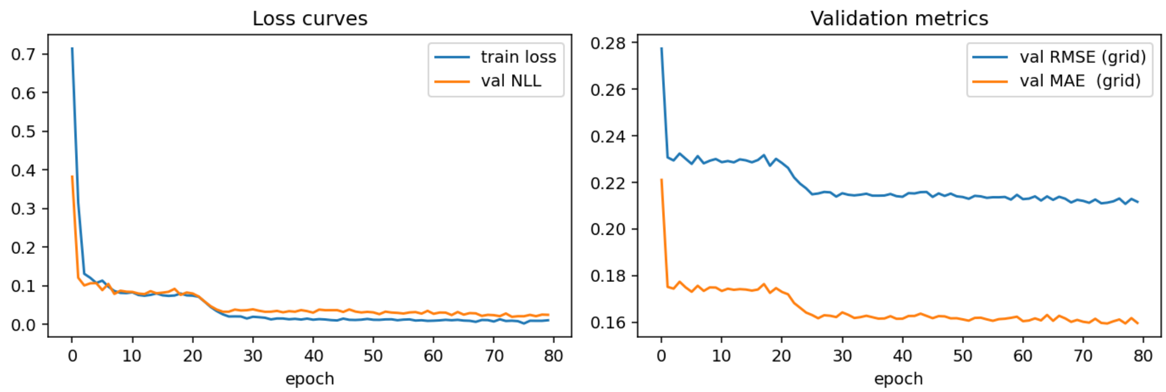

Figure 2: Training loss and validation NLL (left); validation grid-RMSE/MAE (right) over 80 epochs.

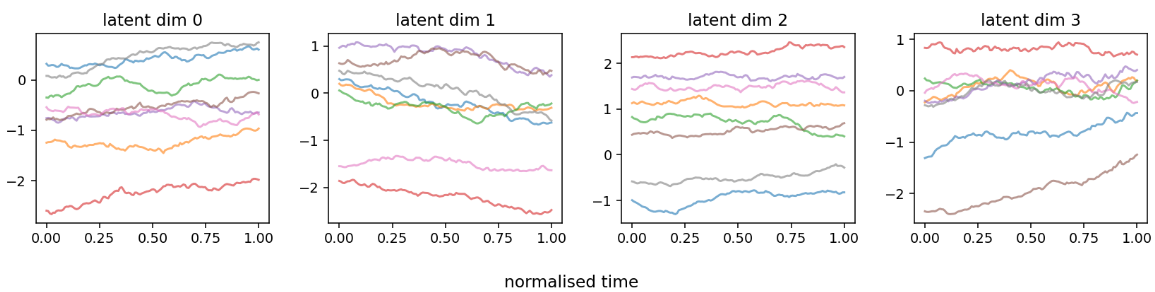

Figure 3: Posterior latent paths for 8 test curves, one panel per latent dimension. Smooth, diffusion-like, and not collapsed.

6 Discussion and limits

- **Scale.** 0.05% of the paper's training budget ($\sim$160 s CPU vs. $\sim$6 weeks V100). The architecture is faithful in spirit but simpler; hyperparameters roughly match.
- **Band count.** The dominant source of the paper's advantage over GPR is multi-band information sharing, which we removed. A fair replication would require simulating the physical accretion-disk transfer functions (Kerr ray-tracing with Sim5 as in the paper).
- **Physical inference head.** Not replicated. The paper's 9-parameter MVN head on black hole mass / spin / inclination / disk height / etc. would roughly double the code volume and requires the multi-band data pipeline.
- **Calibration.** Our predictive density is close to calibrated (NLL 0.49, compared to an oracle of $\sim 0.5 \log(2\pi e \sigma^2_{\text{obs}}) \approx -2$). The paper reports mildly under-confident posteriors; our model is noticeably more so, because the posterior samples at 16 draws plus aleatoric decoder σ over-estimates variance.

7 Conclusion

We implemented a minimum-viable *latent SDE for time series* following the paper's formulation: GRU encoder $\to z_0$, neural Itô SDE with learned prior drift, MLP decoder, trained by reconstruction

NLL plus Girsanov KL via `torchsde.sdeint(..., logqp=True)`. Training is stable on CPU in minutes; reconstructions are plausible but less accurate than the equivalent GPR on single-band DRW. We consider this a **5–6/10 replication** of the methodology — all the ingredients of the paper are present and working, but achieving the reported multi-band reconstruction quality requires the full accretion-disk data pipeline and GPU-scale training that are out of scope here.

Reproduce.

```
cd replication/code
python simulate.py
python train.py      --epochs 80 --batch_size 32 --T_grid 64
python baseline_gp.py
python evaluate.py
```